HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

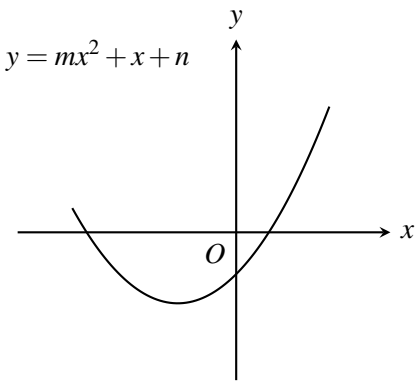
Student No.: 11 Name: _____ Marks: / 32

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The figure shows the graph of $y = mx^2 + x + n$, where m and n are constants. Which of the following is true?



A. $m < 0$ and $n < 0$

B. $m < 0$ and $n > 0$

C. $m > 0$ and $n < 0$

D. $m > 0$ and $n > 0$

(DSE-2014 PII Q05)

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2. If the angle and the radius of a sector are decreased by $x\%$ and 50% respectively so that its area is decreased by 90%, then $x =$

A. 20.

B. 40.

C. 60.

D. 80.

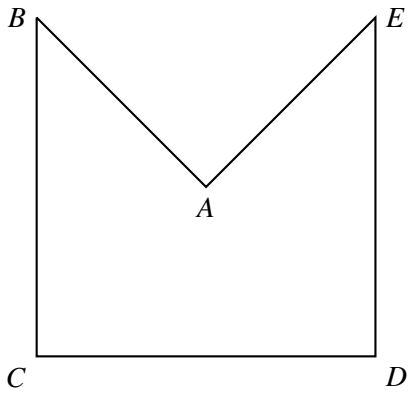
(DSE-2014 PII Q10)

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3. In the figure, $AB = AE$ and $\angle BAE = \angle BCD = \angle CDE = 90°$. If $BC = CD = DE = 16$ cm, then the area of the pentagon $ABCDE$ is



A. 71 cm^2.

B. 128 cm^2.

C. 192 cm^2.

D. 224 cm^2.

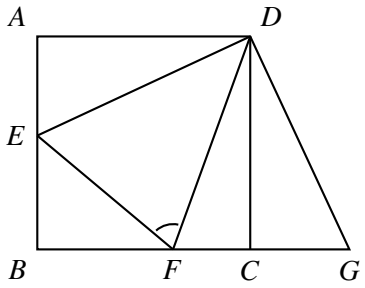
(DSE-2014 PII Q15)

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4. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

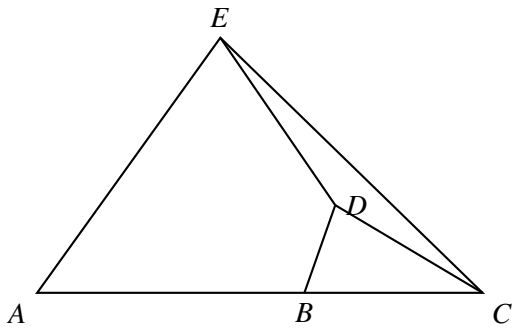
(DSE-2014 PII Q16)

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5. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

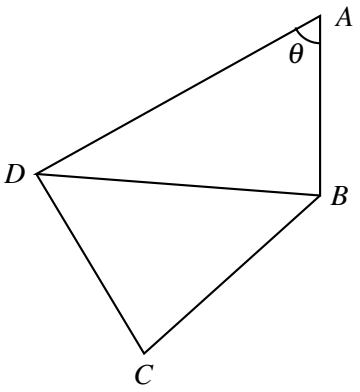
(DSE-2014 PII Q17)

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6. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

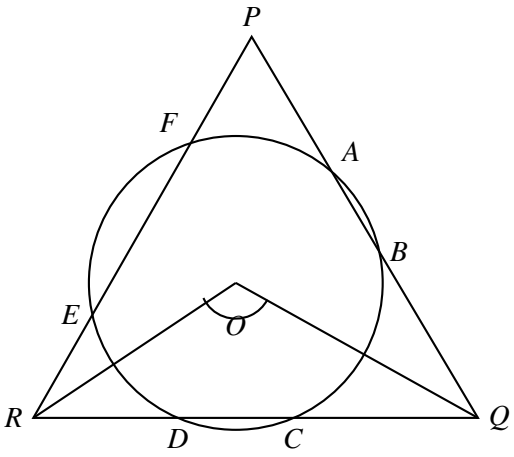
(DSE-2014 PII Q18)

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7. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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8. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

Stem (tens)	Leaf (units)
1	h 4 6
2	3 3 3 4 6 7 7
3	1 2 2 2 6 8
4	0 k

If the range of the above distribution is at least 33, which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2014 PII Q30)

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9. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

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10. Let $f(x) = 3x^2 - 6x + k$, where k is a constant. If the y-coordinate of the vertex of the graph of $y = f(x)$ is 7, then $k =$

A. 1.

B. 3.

C. 4.

D. 10.

(DSE-2014 PII Q35)

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11. If β is a real number, then $\dfrac{\beta^2+4}{\beta+2i} =$

A. $\beta-2i$.

B. $\beta+2i$.

C. $2-\beta i$.

D. $2+\beta i$.

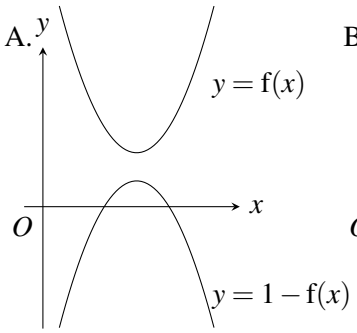
(DSE-2014 PII Q36)

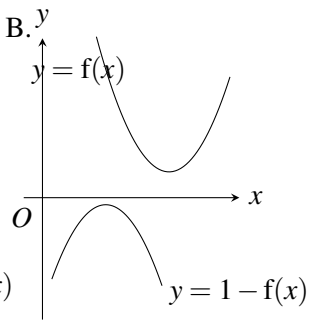
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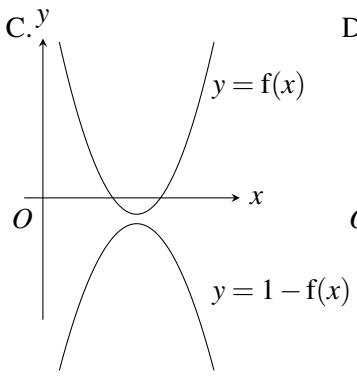
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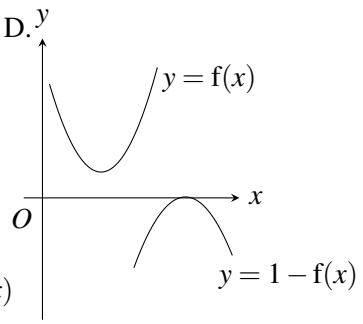
Answers written in the margins will not be marked.

12. Which of the following may represent the graph of $y = f(x)$ and the graph of $y = 1 - f(x)$ on the same rectangular coordinate system?

A.


B.


C.


D.


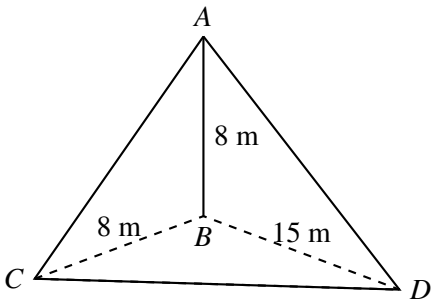
(DSE-2014 PII Q38)

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13. In the figure, AB is a vertical pole standing on the horizontal ground BCD, where $\angle CBD = 90°$. If the angle between the plane ACD and the horizontal ground is θ, then $\tan\theta =$



A. $\frac{8}{15}$.

B. $\frac{15}{8}$.

C. $\frac{15}{17}$.

D. $\frac{17}{15}$.

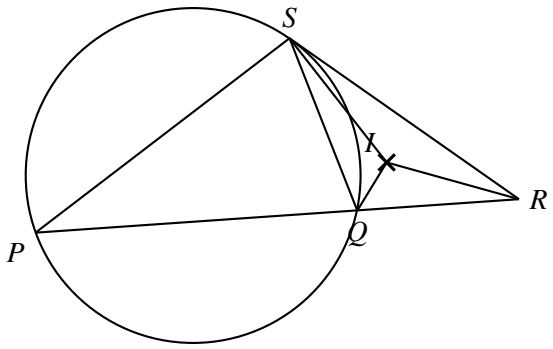
(DSE-2014 PII Q40)

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14. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

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15. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at A and B, then the x-coordinate of the mid-point of AB is

A. $1 + k$.

B. $1 - k$.

C. $\dfrac{1+k}{2}$.

D. $\dfrac{1-k}{2}$.

(DSE-2014 PII Q42)

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16. In an examination, Peter gets 55 marks and his standard score is -3 while Mary gets 95 marks and her standard score is 2. Find the mean of the examination scores.

A. 8 marks

B. 64 marks

C. 75 marks

D. 79 marks

(DSE-2014 PII Q44)

I can finish _____ questions

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